A Cross-Platform Personal Finance App with AI Integration

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Why I Chose this Project

- A passion for personal finance
- Too many lazy advisors
- Robo advising is the future of basic financial planning

Problem Overview

MANY PEOPLE DON'T KNOW IF THEY'RE SAVING ENOUGH

RETIREMENT PLANNING IS CONFUSING

TOOLS ARE FRAGMENTED OR OVERLY COMPLEX

USERS NEED SIMPLE, ACTIONABLE INSIGHTS

Project Goals

01

Build a clean cross-platform UI

02

Implement secure login and user storage

03

Run Monte Carlo retirement simulations on backend

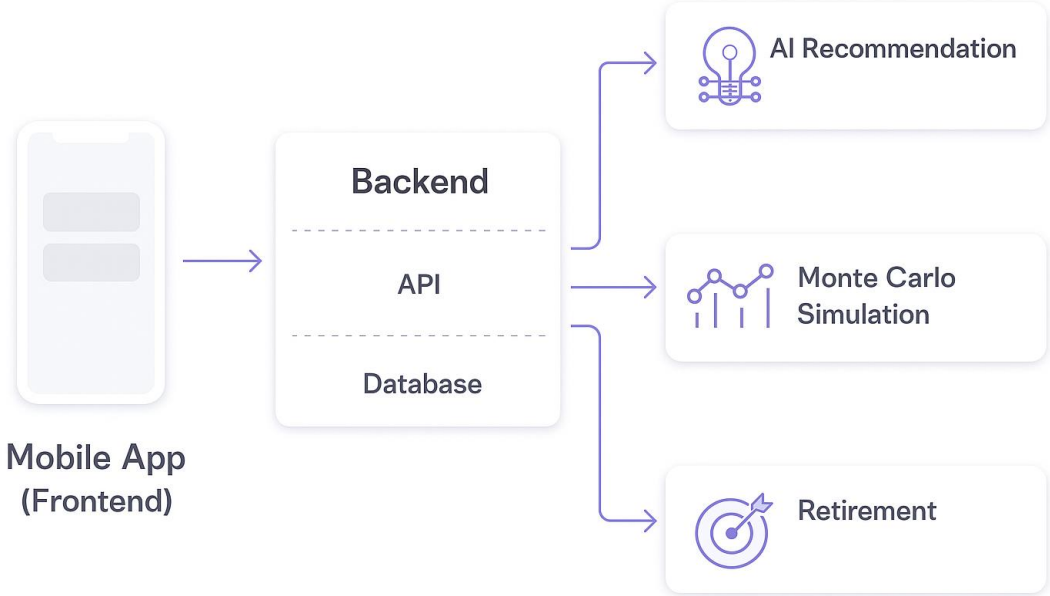
04

Display results visually with an interactive chart

05

Use an AI model to suggest next financial actions

High-Level Architecture



- Client (Kotlin Multiplatform App)
- API (FastAPI)
- Database (PostgreSQL)
- Monte Carlo Engine (Python + NumPy)
- AI Recommender (PyTorch model)

Backend: Authentication & Database

- FastAPI routes: `/register`, `/login`, `/simulate`
- OAuth2 Bearer token
- Password hashing (bcrypt)
- PostgreSQL user storage via SQLAlchem

Monte Carlo Retirement Simulation

Uses asset returns + volatility

Runs thousands of iterations

- Produces:
  - success probability
  - yearly balances
  - ending balance distribution
  - age projections

AI Recommendation Engine

- 100 curated financial recommendations
- Heuristic labeling + PyTorch training
- Takes simulation + user data
- Outputs top 3 personalized suggestions

Kotlin Multiplatform UI

- Compose Multiplatform
- Step-by-step retirement wizard
- Interactive bar/line chart
- Tooltip showing age + balance
- Clean, modern design

Live Demo

Challenges & Solutions

- Cross-platform networking issues
- Serialization mismatches
- Monte Carlo performance
- AI model integration
- UI chart scaling & gestures

Future Improvements

- Add multiple-user household modeling
- Sync data to cloud across devices
- Add deeper recommendation categories
- Portfolio rebalancing suggestions
- Deployment with Docker + CI/CD

Conclusion